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Electric Control Surface Actuator Design Optimization and Allocation for the More Electric Aircraft

Imon Chakraborty,* David Trawick,* David Jackson,*

Dimitri N. Mavris †

Aerospace Systems Design Laboratory, School of Aerospace Engineering

Georgia Institute of Technology, Atlanta, GA, USA

This work pertains to the optimization of electric actuators for the primary and secondary flight control surfaces of a More Electric Aircraft. Electrohydrostatic and electromechanical actuators are considered and optimized in accordance with the flight loads and actuator requirements identified in a separate work by the same authors. For the purposes of this work, the Boeing 737-800 aircraft is chosen as a test case. However the methodology is general and can be applied to any existing design or proposed design concept. Once the optimized actuator designs are obtained, the possibility of using both actuator types by allocating them to control surfaces on the basis of reliability requirements is considered.

I. Introduction

A. The Case for More Electric Control Surface Actuation

Although the concept of electric drive systems for aircraft control surfaces is not a new one,[1] aircraft designers and manufacturers have traditionally and consistently preferred the use of a centralized hydraulic system. Initially in the 1950s, this choice was naturally dictated by the relatively higher maturity of hydraulic technology compared to electric technology at the time. However the continuation of this trend to the present day seems to point to a "better-the-devil-you-know" philosophy.[2] This has led to the maturation of hydraulic system design to a point where, although the systems are adequate to the task and demonstrate the demanded reliability, it is progressively more difficult to obtain performance enhancements. This so called 'technology saturation' of hydraulic systems (and other conventional subsystem solutions), in conjunction with advances in electric drive technology, has led to a renewed and serious consideration of the latter for aircraft systems, in what has come to be called the More Electric Initiative (MEI).

In centralized hydraulic systems, engine driven pumps pressurize hydraulic fluid which acts on hydraulic actuators to move the aircraft control surfaces as required. Safety is assured through redundancy, in which critical components are essentially duplicated, giving rise to functionally and (somewhat) physically separated hydraulic systems. The price that is paid for this redundancy is naturally a weight penalty, increased complexity, and a higher parts count.

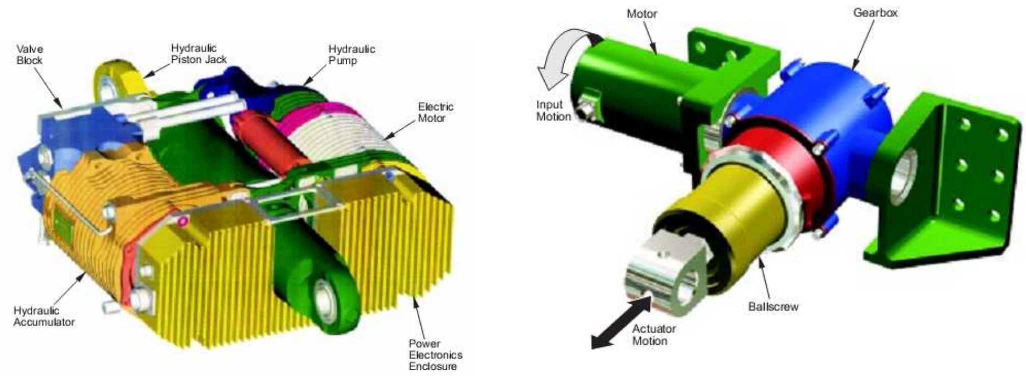
One of the key arguments in favor of electric control surface actuation solutions is the concept of "Power on Demand" or "Pressure on Demand",[2] in which the actuators need to be energized only when a control surface has to be moved or a load has to be overcome. This largely eliminates the continual losses that are present in a centralized hydraulics system. That the elimination of bulky hydraulic lines yields weight savings has already been demonstrated by the design of the Airbus A380 (which uses a 2H/2E arrangement in lieu of the conventional 3H arrangement), where a 1000 lb weight-saving[3,4] has been attributed to the use of electrical actuators. The More Electric Aircraft (MEA) of the future, possessing only electric actuators and no central hydraulics, may enjoy even more weight benefits.

Weight reduction and the associated reduction in operational and maintenance costs are not the only advantages projected for electric control surface actuators. The elimination of hydraulic lines

*Graduate Researcher, Aerospace Systems Design Laboratory, School of Aerospace Engg., Georgia Tech.
†Boeing Chaired Professor, School of Aerospace Engg., and Director, Aerospace Systems Design Laboratory, Georgia Tech.

gives designers added flexibility in architecting other subsystems, and for military aircraft, a reduction in operational vulnerability is also theorized. These projected advantages and the current technical feasibility of electric actuators warrants further investigation into the aircraft-level impact in order to ascertain economic viability.

B. Electrohydrostatic vs. Electromechanical Actuation - an Ongoing Debate



(a) Electrohydrostatic Actuator (EHA) (b) Electromechanical Actuator (EMA)

Figure 1. Typical EHA and EMA layout (courtesy TRW Aeronautical Systems[5])

Electric actuators for control surfaces fall into two main categories depending on how the electric power of the motor is transmitted to the movement of the control surface. In *electrohydrostatic actuators* (EHA), the motor is mated to a pump, which pressurizes and actuates a hydraulic cylinder that is connected to the control surface via a mechanical linkage. In *electromechanical actuators* (EMA), the motor's rotational output is converted to a translational motion via a kinematic converter such as a ball-screw (or a roller-screw). Depending on the motor torque-speed characteristics, a reduction gearbox may or may not be necessary. Figure 1, reproduced from Botten et al.,[5] shows typical design layouts for these two types of actuator units.

Both the EHA and the EMA options have their relative advantages and disadvantages,[5–7] and as such, it is not clear at this point in time which one is the better long-term option for the MEA of the future. It is generally accepted that when sized for the same actuation requirements, the EMA enjoys a weight advantage over the EHA. However, the EMA has the potential for a mechanical jam that has led some to question its applicability especially to primary (flight-critical) control surfaces, and at least one test program experienced over-heating problems.[8] Since almost all current commercial aircraft have a multitude of wing and empennage control surfaces that together comprise the primary and secondary flight control systems, it is quite possible that medium/large-size commercial MEA of the future may use a combination of EHA and EMA units to actuate the various control surfaces, and this possibility will be addressed later in the work.

C. The Proposed Approach

In Chakraborty et al.,[10] we described the development of a tool for estimating aerodynamic loads on flight control surfaces, and also the governing relationships for the components of EHA and EMA designs. In addition, a weight estimation code was developed for these components, and the weight predictions from the tool were calibrated against published weights from previous electric actuator test programs or concept studies. Additional information regarding the calibration is provided in Appendix A.

In this work, the optimal design of these two actuator types for the control surfaces of the Boeing 737-800 (Fig. 2) aircraft will be considered. The maximum control surface hinge moments for the ailerons, elevators, rudder, and spoilers of the Boeing 737-800 were evaluated using a first-order empirical hinge moment evaluation code based on the Roskam method.[11] For the high lift devices, this code was inapplicable, and the loads were estimated based on published aerodynamic data or actuator specifications. The constraining aerodynamic loads for the control surfaces are presented in Tables 1 and 2.

Once the optimal designs are obtained, trends in the design variables as well as the objective function will be analyzed. Then, the actual feasibility of utilizing an EHA or EMA design for a particular control surface will be evaluated, taking into account reliability/redundancy requirements. The potential

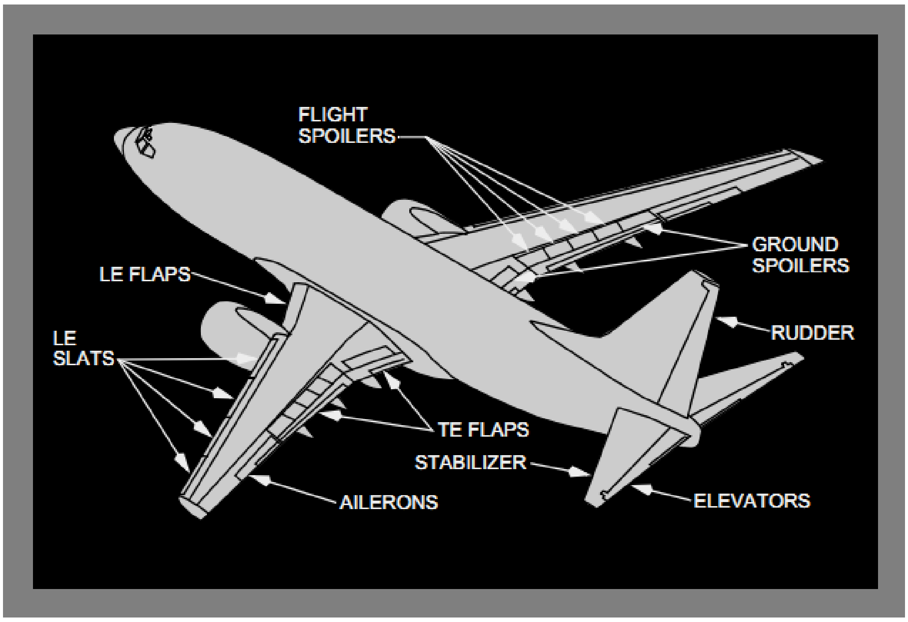


Figure 2. Boeing 737 flight control surfaces[9]

Table 1. Maximum hinge moments and flight conditions for ailerons, elevators, rudder, and spoilers

Control	Max. hinge moment (Nm)	Flight condition
Ailerons	4,200	$V = V_C$, sea-level
Elevators	7,600	$V = V_{Mo}, M = M_{Mo}$
Rudder	8,200	$V = V_{Mo}, M = M_{Mo}$
Flight spoilers	4,200	$V = V_D$, sea-level
Ground spoilers	3,800	$V = 196$ KEAS, sea-level

Table 2. Actuator specifications for high lift system

High-lift device	2-slotted TEF	LE slats	Krueger flaps
Flaps/Aircraft	4	8	4
Actuators/flap	2	1	1
Sizing load	51 kN	6.3 kN	5.6 kNm
Output speed	102 mm/s	60 mm/s	16 deg/s

advantage of a MEA design that uses EHAs for certain control surfaces and EMAs for others will be evaluated.

The rest of this paper is organized as follows. Section II will deal with the formulation of the constrained optimization problem whose solution will yield a feasible and optimal design. Section III will present the optimized EHA and EMA designs for the flight control surfaces. Section IV will consider the case of allocating EHA and EMA units to different control surface types based on the characteristics and trends seen from the all-EHA and all-EMA designs. Section V will conclude with a summary of the results and proposals for future work and extensions.

II. The Constrained Design Optimization Problem

The design of the individual actuator units will be driven by the solution of a constrained optimization problem of the following form:

$$\begin{array}{rcll} \text{Determine} & \bar{x} & = [x_1, x_2, \ldots, x_n] & \text{(Design vector)} \\ \text{to minimize} & f(\bar{x}) & = f(x_1, x_2, \ldots, x_n) & \text{(Objective function)} \\ \text{subject to} & g_j(\bar{x}) & \leq 0, \quad j = 1, \ell & \text{(Inequality constraints)} \\ & h_k(\bar{x}) & = 0, \quad k = 1, m & \text{(Equality constraints)} \\ x_i^L \leq & x_i & \leq x_i^U \quad i = 1, n & \text{(Bound constraints)} \end{array} \tag{1}$$

In Hegde et al.,[12] the authors solved an optimization problem of this form using surrogate model based optimization. In this work however, the dimensionality of the design space for each actuator is reduced by considering a smaller number of design variables and internally satisfying certain constraints during the actuator sizing. As a result, it was possible to directly use a gradient-based optimization scheme.

A. Choice of Design Variables ($\bar{x}$)

Blanding[7] makes a very intersting observation regarding the sizing of hydraulic actuators, which explains why such actuators are typically oversized for their applications. The actuator cylinder area is typically sized based on a consideration of the "corner horsepower ", which is the product of the maximum control surface hinge moment with the maximum control surface angular rate. Now, the stroke of the actuator is directly dependent upon the control surface angular movement range, which in turn is determined by low speed control effectiveness. The maximum hinge moment does not occur in this condition. Further, the maximum surface rate is typically set at yet another flight condition. This situation, where the actuator bore, stroke, and rate are sized by three different flight conditions which typically do not coincide results in a design that has a higher horsepower than is required for the application. Blanding argues that an EMA (and in fact, even an EHA) does not need to deliver the same power as the hydraulic actuator that it replaces. It only needs to have sufficient torque output (to satisfy the maximum hinge moment required) and sufficient speed (to satisfy the maximum angular rate requirement). Taking this argument into account, and considering the design principles of the two actuators types, the design variables shown in Table 3 were selected.

Table 3. EHA and EMA Design variables

Variable Name	Symbol	Unit	EHA	EMA	Description
Kinematic gearing	G_k	rad/m	X	X	Control angular range / actuator stroke
Ballscrew gearing	G_{bs}	mm/rev		X	Ballscrew kinematic characteristic
Gearbox ratio	G_{gb}	–		X	Reduction ratio of EMA gearbox
Pump displacement	D_p	mm^3/rev	X		Theoretical volume pumped / revolution
Pump max pressure	Δp_{max}	MPa	X		Max. pump operating pressure diff.

B. Candidate Objective Functions ($f(\bar{x})$)

The most commonly cited benefits for electric actuators are weight savings and increased energy efficiency, and so clearly the weight of the sized actuator and its energy consumption are two important metrics. However, regarding the energy consumption, it must be remembered that a future Energy Optimized Aircraft (EOA) or Power Optimized Aircraft (POA) will require a consideration of energy consumption at the vehicle (or system) level, not merely at the subsystem level. Thus, it would not be appropriate to energy-optimize the control actuation subsystem in isolation without considering the simultaneous energy-optimization of other key aircraft subsystems.

Actuator weight on the other hand can be considered at the unit level, and thus has been used here as an objective function, i.e. minimize actuator weight. The weight of an actuator unit (be it an EHA or an EMA) can be built up from estimates of the weights of its components, as shown in Eq. 2, with the quantity $W_{misc.}$ accounting for the weights of fixtures, etc. whose weights are difficult to accurately determine at the conceptual design level, and which may be represented as a reasonable fraction of the weight of the major components. Details regarding the weight estimation are available in Chakraborty et al.[10]

$$\begin{aligned} W_{EHA} &= W_{cyl} + W_{pump} + W_{motor} + W_{misc.} \\ W_{EMA} &= W_{ballscrew} + W_{gearbox} + W_{motor} + W_{misc.} \end{aligned} \quad (2)$$

The number of actuators driving a control surface is determined based on the redundancy philosophy, and takes into consideration the criticality of the control surfaces toward the safe continuation of the flight following a failure, failure probabilities, etc. For current hydraulic systems[13] and in proposed EHA systems,[14] two to three actuators are used in tandem for flight-critical control surfaces such as ailerons, elevators, and rudder, while surfaces that are less critical, such as spoilers, are typically actuated by a single actuator per surface. The same trends will be adhered to in this work. Elements of the high-lift system such as slats and Krueger flaps will be actuated by a single actuator per flap panel (as is done in reality for the Boeing 737-800). Each double-slotted trailing edge flap will be driven by two actuators positioned at its spanwise extremities (in reality the Boeing 737-800 trailing edge flap system is centrally powered by a Power Distribution Unit (PDU) and left and right side flap panels are synchronized mechanically using torque tubes / shafts).

If multiple actuators are used in tandem for a single control surface in an active-active arrangement, then there is the possibility of down-sizing each actuator, which results in weight savings.[13] However, in case an active-standby arrangement is used, each actuator must individually be rated to take up the entire actuation load. In order to factor in both of these design philosophies, the current work will show optimized actuators that are rated to take up 100%, 70% and 50% of the control surface actuation load for control surfaces that are driven by more than one actuator.

It should be noted that the reliability criteria are not considered in the design optimization phase of the work. In this phase, both EHA and EMA concepts will be optimized for each control surface based on the estimated control surface actuation load and the percentage of said load to be taken up by the actuator (100%, 70% or 50%). The allocation phase of the study will consider the feasibility of allocating each actuator type to the various control surfaces of the aircraft.

C. Definition of Constraints ($g_j(\bar{x}), h_k(\bar{x})$)

Table 4 shows the constraints that must be satisfied by a candidate design for it to be considered feasible. Note that each constraint shown in Table 4 must be cast in the form $g_j(\bar{x}) \leq 0$ for the optimizer. For the EHA, the cylinder and piston are sized based on the selected pump maximum pressure, which is a design variable for the optimizer. Once the piston mass is known, the Δp_{max} constraint ensures that the available maximum pressure difference is sufficient to accelerate the piston against the external load with a suitable acceleration margin available. For control surfaces having a high aerodynamic load *and* a high angular rate, the 10,000 RPM / 50 A constraint thresholds were applied, while for control surfaces that had a lower velocity requirement were optimized using the 5,000 RPM / 30 A constraint thresholds.

For the thermal constraint, the motor winding temperature was constrained to not exceed 160°C against a local ambient temperature that may be as high as 70°C.[15] For the aileron, elevator, and rudder actuators, two thermal cases were considered. In the first, the actuator was required to hold the surface indefinitely against the steady-holding load, and the steady-state temperature was noted. In the second case, the actuator was required to output nominally the steady-holding load, while control surface deflections that increased the load up to the stall load were then superimposed. This represents an extreme operating case where the actuator must output the steady-holding load to hold the control surface at a deflected position (e.g., rudder, following failure of wing-mounted engine), while rapid movements of the control surface are required on top of that in order to exercise positive control over the aircraft.

For the flight spoilers, the constraining thermal case was assumed to be an emergency descent from altitude with the flight spoilers deployed in the "speedbrake" function.[16] In this case, since the descent begins from high altitude, the motor ambient temperature was set based on the outside atmospheric temperature as a function of altitude using an atmosphere model.

For the high lift devices (double-slotted trailing-edge flaps, and leading edge Krueger flaps and slats) and ground spoilers, the constraining thermal case, as pointed out by Bennett[15,17] is one where the aircraft performs repeated takeoffs and landings for the purpose of flight training. This naturally subjects the high lift system to a much more demanding duty cycle than would be encountered in a routine point-to-point flight (where deployment is limited to the takeoff and/or landing phases).

Table 4. **Design constraints**

Parameter	Constraint	EHA	EMA
Required Δp_{max}	< pump rated Δp_{max}	X	
Ballscrew RPM	< 2,000 RPM		X
Motor max. RPM	< 5,000 RPM / < 10,000 RPM	X	X
Motor stall current	< 30 A / < 50 A	X	X
Motor winding temperature	< 160°C	X	X

III. Overview of Optimized Actuator Designs

Since the objective function for the optimization was chosen to be the actuator weight, this section compares the weight trends for the various control surface actuators. For continuity of presentation, only the weights of the optimized actuators are presented in this section, and additional details regarding the designs are deferred to Appendix B.

Table 5 shows the weights of the EHA and EMA designs for the three load percentages chosen for the aileron, elevator, and rudder actuators. For these three control surfaces, the EHA configuration that was used featured two electrohydraulic channels (i.e., two motors, each mated to a pump) driving a tandem cylinder. This arrangement was shown by Sadeghi et al.[14] to provide adequate reliability for a flight-critical control surface. For the EMA, two electromechanical channels (i.e., two motors, each mated to a gearbox) were used to drive a single ballscrew through a summing differential. This arrangement had been used for the C-141 Starlifter electric aileron actuator test program (Thompson[18]).

It is seen that for each surface and for each load percentage, the optimal EMA design weighs less than the optimal EHA design, a trend that is often claimed to be an advantage of the EMA concept. In order to investigate the magnitude of this weight advantage, the ratio of EHA weight to EMA weight is plotted as a function of the load percentage in Fig. 3 for the three control surface types. It is evident from this plot that the weight advantage of the EMA is maximum when the control surface hinge moment is high (elevator and rudder) and when the actuator is sized for a higher percentage of that hinge moment.

Table 5. **Weight-optimized EHA and EMA for aileron, elevator, and rudder**

Actuator type → Load % → Control Surface	EHA (100%) (kg)	EHA (70%) (kg)	EHA (50%) (kg)	EMA (100%) (kg)	EMA (70%) (kg)	EMA (50%) (kg)
Aileron	22.2	16.3	12.5	18.6	14.3	11.2
Elevator	37.3	27.2	20.2	26.7	20.3	16.5
Rudder	40.0	28.7	21.3	29.0	22.5	18.1

Table 6. **Weight-optimized EHA and EMA for flight and ground spoilers**

Actuator type → Configuration → Control Surface	EHA 1EH, SC (kg)	EHA 2EH, TC (kg)	EMA 1EM, 1BS (kg)	EMA 2EM, 1BS (kg)
Flight spoiler	11.5	22.2	9.5	17.7
Ground spoiler	9.2	17.2	8.2	13.9

Table 6 shows the weight-optimal EHA and EMA designs for the flight and ground spoilers. For each actuator type, two configurations are shown. For the EHA, the first configuration (1EH, SC) features a single electrohydraulic channel (i.e., 1 motor mated to 1 pump) driving a single cylinder. The second configuration (2EH, TC) features two electrohydraulic channels (2 motors, each mated to 1 pump) driving a tandem cylinder. For the EMA, the first configuration (1EM, 1BS) features a single electromechanical channel (1 motor mated to 1 gearbox) driving a ballscrew, while the second configuration (2EM, 1BS) features two electromechanical channels (2 motors, each driving a gearbox) driving a single ballscrew. The weight trends seen for the spoilers is similar to those seen for the primary flight control surfaces.

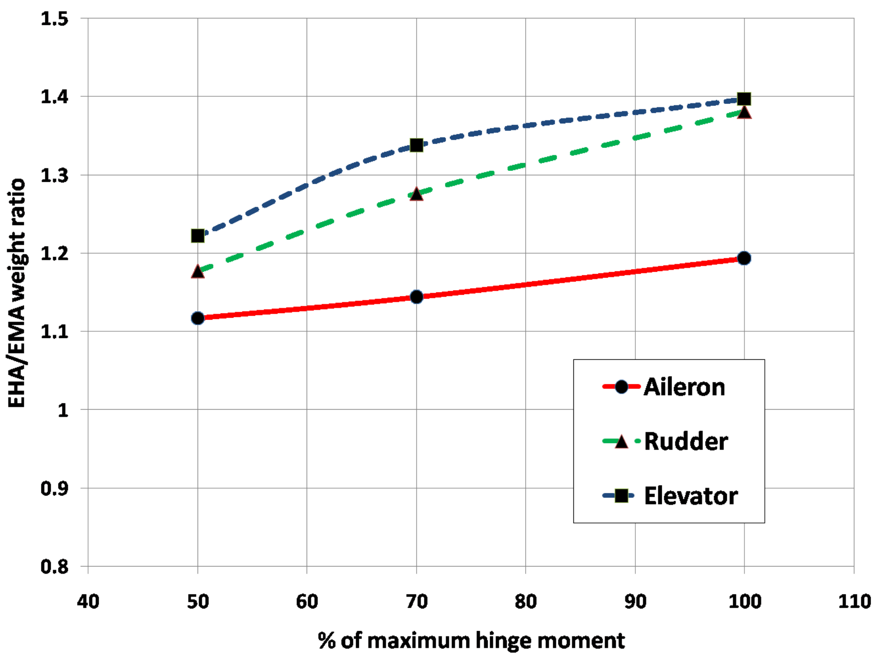


Figure 3. Ratio of EHA weight to EMA weight

Table 7. Weight-optimized EHA and EMA for Krueger flaps and slats

Actuator type →	EHA	EHA	EMA	EMA
Configuration →	1EH, SC	2EH, TC	1EM, 1BS	2EM, 1BS
Control Surface	(kg)	(kg)	(kg)	(kg)
Krueger flap	18.2	32.8	11.7	21.0
Leading edge slat	16.0	27.4	10.6	18.0

Table 7 contains the weights of the optimized EHA and EMA designs for the leading-edge devices - the Krueger flaps and the slats. Even though the actuation load for the slat is much lower than that for the Krueger flap, the actuator weights of both are similar. An analysis of the weight breakdown (not shown here for brevity) showed that this was down to the longer stroke needed for the three-position slat.

For the trailing-edge flap, only electromechanical actuation was considered in this work. In keeping with the current deployment mechanism for the TEFs of the Boeing 737-800, the EMA configuration used is different from that used for the other control surfaces, and features a rotating screw and a translating nut (the EMAs for the other controls use a rotating nut and a translating screw). Since the flap panel has a considerable spanwise extent, it must be driven at both ends to prevent any skewing of the panel under aerodynamic loading, and for this purpose two EMA units per flap panel were considered, and the weights of the optimal designs are presented in Table 8.

Table 8. Weight-optimized EMA for trailing-edge flaps

Actuator type →	EMA (2EM/1BS)		
Load % →	(100%)	(70%)	(50%)
Control Surface	(kg)	(kg)	(kg)
TEF	50.6	40.6	33.6

IV. Actuator Allocation in a Hybrid EHA/EMA Design

In the previous section, weight-optimized actuator designs of both EHA and EMA type have been presented for the various control surfaces of the Boeing 737-800 aircraft. At that stage, the actuators were sized and optimized based solely on the control surface load and rate requirements. The analysis presented in this section deals with the feasibility of allocating these two actuator types to the control surfaces taking into account reliability requirements.

FAR 25.1309 requires that any combination of failures with catastrophic consequences must be shown to be Extremely Improbable, which quantitatively translates to a failure probability of less than 10^{-9} (sometimes an even more stringent 10^{-10}) per flight hour.[19] Dealing specifically with flight controls, FAR 25.671 requires (at least) that no single failure be sufficient to bring about a catastrophic consequence.

A relevant subpart of FAR 25.671 is the one that deals specifically with the issue of a control surface jam (since this is a risk for the EMA design). It states that the aircraft should remain capable of continued safe flight and landing after

"Any jam in a control position normally encountered during takeoff, climb, cruise, normal turns, descent, and landing unless the jam is shown to be extremely improbable, or can be alleviated. A runaway of a flight control to an adverse position and jam must be accounted for if such runaway and subsequent jamming is not extremely improbable."[20]

Flight critical control surfaces (ailerons, elevators, rudder) driven with multiple EHA units can satisfy this requirement. As demonstrated by Sadeghi et al.,[14] two units per control surface, with each unit containing dual electrohydraulic channels, and redundant electronics, can satisfy the 10^{-9} failure probability requirement. Moreoever, as a nature of their design (incorporating the hydraulic cylinder), EHA units can be made to revert to a damped mode following a failure. For a control surface driven by only one EHA unit (such as a spoiler), this makes it possible for the control surface to "blow back" under the dynamic pressure of the freestream to a faired position where it contributes little to no adverse moment on the aircraft, and following which the inherent damping prevents control surface flutter. For control surfaces driven by multiple EHA units, it is possible for the functioning actuator(s) to drive the control surface (after overcoming the damping resistance of the failed-but-movable EHA).

For the EMA, the situation is complicated by the fact that for the general EMA layout, the ballscrew remains a single point of failure for the unit. This means that despite the redundancy that can be built in *upstream* of the ballscrew (using redundant actuator electronics and dual electromechanical channels), the overall failure probability of the unit is still largely driven by the failure probability of the ballscrew (Fault trees corresponding to some EMA layouts is presented in Appendix C). Further, in case of an EMA jam, it is not possible to backdrive the actuator. This means that a control surface driven by a single EMA will not blow back under freestream dynamic pressure, and if a control surface is driven by multiple EMAs, it will not be possible for the functioning EMA to drive the control surface after overcoming the jammed EMA. If this is the case, then it would be impractical to use multiple (say two) EMAs on a control surface, since the overall probability of the control surface jamming (through either of the EMA units jamming) would essentially be doubled.

EMA concepts have been proposed that feature clutch mechanisms that allow a failed/jammed EMA to be effectively uncoupled from the control surface that it is driving, thereby allowing blowback or continued actuation by a parallel (and functioning) EMA actuator. In addition, "fail-safe" ballscrews[21] have been proposed (for high-lift systems) which may allow the EMA to have an overall failure probability more comparable to the EHA. These enhancements will likely add to the weight of the EMA unit, and thus partially negate one of its advantages over the EHA design. At present, the authors do not have an estimate of the magnitude of the weight gain, but if the possibility of a control surface jam can be alleviated, then it may be possible to enjoy at least some weight savings by the use of the EMA for primary flight controls.

EMAs can also be considered for the non flight-critical control surfaces, such as spoilers.[16,22] The Boeing 737-800 has four ground spoilers (two on each wing), which are deactivated unless the aircraft main landing gear strut is compressed. These spoilers are not used in flight, and when deployed, always move to their maximum deflection. EMAs can definitely be considered for such types of spoilers, although Table 6 indicates that due to the low control surface actuation load in this case, using an EMA does not give much of a weight advantage over using an EHA design.

The remaining eight spoilers of the Boeing 737-800 (four on each wing) are flight spoilers, which deflect simultaneously on both wings to serve the speedbrake function, and only on a single wing to augment the roll control authority of the ailerons. As pointed out by Fronista and Bradbury,[22] it is possible to incorporate a design feature that prevents the spoiler from deploying any further in the event that the EMA driving it loses electrical power. However, since the runaway and subsequent jamming probability of an EMA cannot be shown to be Extremely Improbable at this stage, FAR 25.671 requires

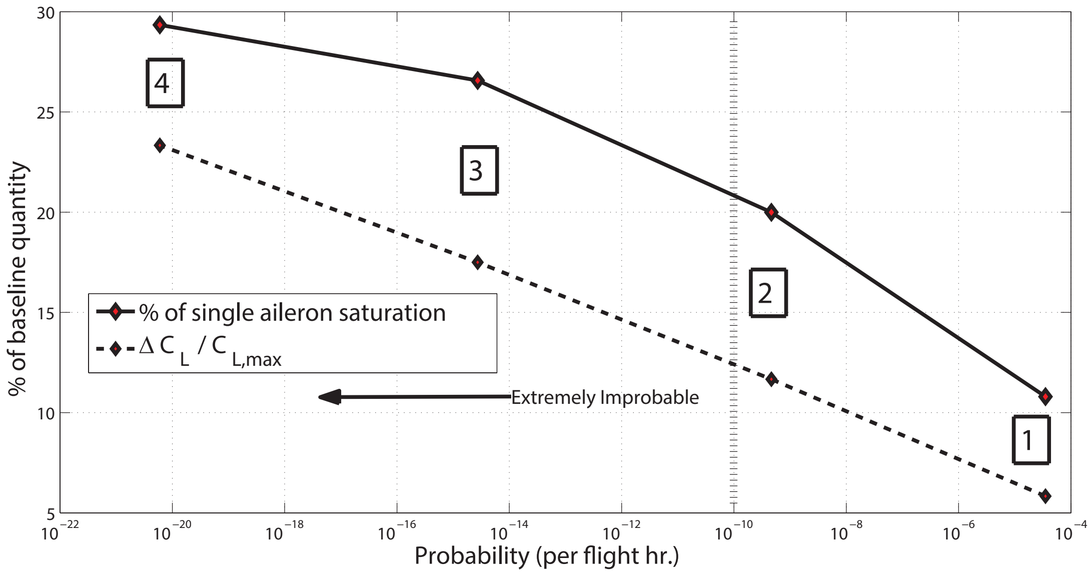


Figure 4. Rolling moment and loss of lift due to one or more spoiler failures

that the controllability of the aircraft following such a runaway and jam be demonstrated. The effect of a deflected (and jammed) spoiler on one wing will be a rolling moment (wing-down for the wing containing the deflected spoiler), a yawing moment (towards the side of the deflected spoiler), a loss of lift, and an increase in drag. Taking into account the geometry of the Boeing 737-800, and assuming (for the purpose of a conservative estimate) that spoilers fail from the outboard side of the wing, and that a failure involves said spoiler(s) always stuck at their maximum deflections, Roskam relationships were used to estimate the rolling moment generated by these spoiler failures as a percentage of the roll authority of a single aileron. Simultaneously, the loss of lift as a percentage of the aircraft clean configuration $C_{L,max}$ was also computed. These quantities were plotted against the probability of their occurrence, as shown in Figure 4. Compound events (i.e. with more than one spoiler failing) occurring to the left of the 10^{-9} or 10^{-10} vertical line may be taken to be Extremely Improbable. Based on a 10^{-10} requirement, it is seen that the simultaneous failure of two spoilers is already close to being an Extremely Improbable event (and the failure of more than two spoilers definitely is), and even assuming that these two failed spoilers are the most outboard ones, the degree of saturation of the ailerons in order to counteract the rolling moment imbalance is not excessive (note: The plot pertains to the roll authority of a *single* aileron, whereas in practice, both ailerons would be used to restore rolling moment equilibrium). A similar result (not shown here) was found for the yawing moment imbalance, and the sustainability of flight given the loss of lift and increment in drag would depend on the available excess power (a multiple spoiler failure at high altitude may necessitate a descent to lower altitude where thrust available is higher). These results seem to indicate that the aircraft would remain controllable after two "worst-case" spoiler failures, and that failures of more than two spoilers is an Extremely Improbable compound event. In this case, it may be feasible to replace the two inboard flight spoilers, or perhaps all four flight spoilers, with EMA units instead of EHA units, thus leading to a weight benefit.

The authors would like to caution that the above statement is made following only a preliminary assessment of the effect of a jammed spoiler. Further, this assessment is valid for the aircraft in question (Boeing 737-800), and for an aircraft using spoilers of larger spanwise and chordwise dimension, and located further outboard on the wing, this conclusion regarding controllability may not be valid.

EMAs may also be applicable to the high-lift system of the aircraft. Bennett et al.[15,17,23–25] consider the use of an electromechanical actuation system for the trailing-edge high lift devices of an Airbus A320 aircraft. Since the high-lift system is not necessarily crucial for the safe continuation of the flight, the probability of an inability to extend or retract the flaps has to be shown to be less than 10^{-5}, which the authors demonstrate is achievable with currently accepted reliability figures for EMA designs provided that a dual-lane electric drive is utilized. However, since a flap asymmetry (disagree) caused by the runaway of a flap on one wing may still lead to a catastrophic occurrence, the probability of loss of control (PLOC) has to be shown to be less than 10^{-10}. Bennett et al. then demonstrate this figure with power-off brakes and multiply redundant control signals and power supplies. While it is impossible for us at this stage to determine the weight increase that would result from this redundancy requirement, the optimal EMA design for the trailing-edge flaps has a weight comparable to that of the current Boeing

737-800 linear ballscrew actuator,[26] and it is likely that some weight benefits could be derived from the elimination of the centralized PDU, torque-tubes, shafts, etc.

Similarly, there is the possibility of using EMAs for the leading edge devices (Krueger flaps and slats). For the Boeing 737-800, each of the leading edge flaps is actuated by a single hydraulic cylinder, and synchronization between left and right side flaps is achieved through electronic sensing rather than mechanically. As with the trailing-edge devices, it may be possible to use EMAs for the leading-edge devices provided power-off brakes and sensors be provided to prevent a runaway (loss of control) scenario.

In order to analyze the magnitude of the weight advantage that a "hybrid" design (using both EHA and EMA units) might enjoy, a baseline "all-EHA" design was considered, that uses EHA units for actuating all control surfaces (ailerons, elevators, rudder, flight spoilers, ground spoilers, Krueger flaps, and slats) except the trailing-edge flaps (which use electromechanical actuation). Next the hybrid design was considered, still utilizing EHAs for the critical flight control surfaces, but EMA units for the flight spoilers, ground spoilers, Krueger flaps, and slats. The weight buildup of these two configurations is shown in Table 9. From this table it is seen that the hybrid design weighs about 9% less than the all-EHA baseline. However, it should be noted that the current work does not take into account the additional weight of power-off brakes and sensing and monitoring equipment that would be required for the EMA actuated surfaces to prevent any uncommanded deployments, runaways, or loss of actuator control. As a result, the actual weight benefit from using the hybrid actuator layout may be somewhat less pronounced.

Table 9. Configuration comparison

Control surface			Config. 1 Weight (kg)			Config. 2 Weight (kg)		
Name	Number	Actuators	Design	Unit	Group	Design	Unit	Group
Aileron	2	2	EHA	16.3	65.3	EHA	16.3	65.3
Elevator	2	2	EHA	27.2	108.8	EHA	27.2	108.8
Rudder	1	3	EHA	28.7	86.2	EHA	28.7	86.2
Flight spoiler	8	1	EHA	11.5	92.0	EMA	9.5	76.1
Ground spoiler	4	1	EHA	9.2	36.6	EMA	8.2	32.8
Krueger flap	4	1	EHA	18.2	72.8	EMA	11.7	46.8
LE slat	8	1	EHA	16.0	127.8	EMA	10.6	84.7
TE flap	4	2	EMA	50.6	404.8	EMA	50.6	404.8
			Total →		994.4	Total →		905.5

V. Conclusion

A MATLAB/Simulink based sizing, simulation, and optimization tool was developed and used to optimize Electrohydrostatic Actuator (EHA) and Electromechanical Actuator (EMA) designs for the primary and secondary flight control surfaces of the Boeing 737-800 aircraft. The actuators were sized and optimized based on the relevant control surface aerodynamic loads that were either internally computed by the tool using an empirical estimation method or (in the case of high lift devices) estimated based on public domain information for the relevant device type. It was observed that for each control surface, the weight-optimal EMA design was lighter than the corresponding weight-optimal EHA design. This weight advantage enjoyed by the EMA was seen to be more pronounced when control surface aerodynamic loads were higher, and when each actuator was sized to take up a greater percentage of the total control surface actuation load.

However, given the single-point failure possibility of the EMA through a mechanical jam, multiple EHAs were allocated to each of the flight-critical control surfaces (ailerons, elevators, and rudder) for both configurations considered, in order to satisfy the much more stringent reliability requirements that must be demonstrated for these control surfaces in order to make a catastrophic occurrence Extremely Improbable. EMA designs were instead considered for the spoilers (both flight and ground spoilers) and the high-lift devices (trailing edge flaps, Krueger flaps, and leading edge slats). This hybrid configuration (featuring EHAs for the critical control surfaces, and EMAs for the others) was compared against a baseline all-EHA configuration, and a weight advantage of around 9% was predicted for the hybrid configuration. Finally, it must be noted that as the component reliability increases further with technology maturation, the optimal actuator allocation may change and will need to be re-evaluated accordingly.

Appendix

A. Weight estimate calibration summary

Table 10. EHA Weight Estimate - Calibration

Program	F-18 SRA flaperon	Typical tandem EHA
Source	Navarro[27]	Sadeghi & Lyons[14]
Stall load (lbf)	13,300	32,000
No load rate (in/s)	7.7	8.0
Stroke (in)	4.5	-
Architecture	1 3ϕ PMM $\rightarrow$ 1 FDP	Dual EH channel, 2 TC/surface
Published weight (lb)	41.5	159.5
Predicted weight (lb)	42	164

Table 11. EMA Weight Estimate - Calibration

Program	F-18 SRA flaperon	Transport spoiler	C-141 aileron
Source	Jensen et al.[8]	Fronista & Bradbury[22]	Thompson[18]
Stall load (lbf)	13,200	50,000	19,050
No load rate (in/s)	6.7	7.0	4.65
Stroke (in)	4.125	6.0	5.43
Architecture	2 3ϕ BLDCM $\rightarrow$ 1 BS	1 5ϕ SRM $\rightarrow$ 1 BS	2 M/GB $\rightarrow$ 1 BS
Published weight (lb)	26	39	35
Predicted weight (lb)	26	40	37

B. Actuator designs

Table 12. Aileron actuator

Actuator type →		EHA			EMA		
Load percentage →		50%	70%	100%	50%	70%	100%
Design variables							
Kinematic gearing	(rad/m)	7.7	7.1	7.5	6.7	6.8	8.0
Pump max. pressure	(MPa)	35	35	35			
Pump displacement	(mm^3/rev)	315	440	630			
Ballscrew gearing	(mm/rev)				12.8	15.7	24.8
Gearbox Gearing	(-)				15.5	25.5	47.3
Actuator characteristics							
Weight	(kg)	12.5	16.3	22.2	11.2	14.3	18.6
Stall load	(kN)	20.3	26.2	39.5	17.6	25.0	42.0
Ram speed	(mm/s)	135	147	140	156	154	131
Stroke	(mm)	90	98	93	104	103	87
Motor characteristics							
Power rating	(kW)	3.1	4.3	6.2	2.7	3.7	5.3
Corner power	(kW)	10.6	14.8	21.0	9.1	12.7	18.0
Peak torque	(Nm)	6.6	9.2	13.4	7.6	8.1	11.4
Continuous torque	(Nm)	2.6	3.6	5.3	3.0	3.2	4.5

Table 13. Elevator actuator

Actuator type →		EHA			EMA		
Load percentage →		50%	70%	100%	50%	70%	100%
Design variables							
Kinematic gearing	(rad/m)	6.3	5.0	6.1	7.6	7.7	6.8
Pump max. pressure	(MPa)	35	35	35			
Pump displacement	(mm^3/rev)	565	790	1130			
Ballscrew gearing	(mm/rev)				7.9	14.5	17.5
Gearbox gearing	(-)				14.3	25.5	25.5
Actuator characteristics							
Weight	(kg)	20.2	27.2	37.3	16.5	20.3	26.3
Stall load	(kN)	29.5	33.0	57.5	36.0	51.1	64.0
Ram speed	(mm/s)	168	209	172	137	135	154
Stroke	(mm)	112	140	115	92	90	103
Motor characteristics							
Power rating	(kW)	5.6	7.8	11.1	5.0	6.8	9.6
Corner power	(kW)	18.9	26.5	37.8	16.9	23.1	32.8
Peak torque	(Nm)	11.7	16.9	23.5	10.8	15.5	23.2
Continuous torque	(Nm)	4.6	6.6	9.2	4.2	6.1	9.1

Table 14. **Rudder actuator**

Actuator type →		EHA			EMA		
Load percentage →		50%	70%	100%	50%	70%	100%
Design variables							
Kinematic gearing	(rad/m)	6.0	8.7	8.2	6.6	6.3	6.7
Pump max. pressure	(MPa)	35	35	35			
Pump displacement	(mm^3/rev)	610	855	1220			
Ballscrew gearing	(mm/rev)				11.4	20.3	25.1
Gearbox gearing	(-)				14.5	30.8	40.5
Actuator characteristics							
Weight	(kg)	21.3	28.7	40.0	18.1	22.5	28.9
Stall load	(kN)	30.4	62.4	83.0	33.8	45.3	68.8
Ram speed	(mm/s)	176	120	130	158	165	155
Stroke	(mm)	106	72	77	95	99	93
Motor characteristics							
Power rating	(kW)	6.0	8.4	12.0	5.2	7.2	10.3
Corner power	(kW)	20.4	28.6	41.0	17.8	24.6	35.1
Peak torque	(Nm)	12.7	17.8	26.0	14.2	15.7	22.3
Continuous torque	(Nm)	5.0	7.0	10.2	5.6	6.1	8.8

Table 15. **Flight spoiler actuator**

Actuator type →		EHA		EMA	
Configuration →		1EH/SC	2EH/TC	1EM/1BS	2EM/1BS
Design variables					
Kinematic gearing	(rad/m)	7.1	7.1	6.3	6.5
Pump max. pressure	(MPa)	35	35		
Pump displacement	(mm^3/rev)	645	645		
Ballscrew gearing	(mm/rev)			19.5	18.2
Gearbox gearing	(-)			29.0	25.5
Actuator characteristics					
Weight	(kg)	11.5	22.2	9.5	17.7
Stall load	(kN)	37.5	37.5	33.1	34.3
Ram speed	(mm/s)	147	147	166	160
Stroke	(mm)	93	93	105	102
Motor characteristics					
Power rating	(kW)	6.2	6.2	5.3	5.3
Corner power	(kW)	21.0	21.0	18.0	18.0
Peak torque	(Nm)	13.4	13.4	11.6	12.6
Continuous torque	(Nm)	5.3	5.3	4.6	5.0

Table 16. Ground spoiler actuator

Actuator type →		EHA		EMA	
Configuration →		1EH/SC	2EH/TC	1EM/1BS	2EM/1BS
Design variables					
Kinematic gearing	(rad/m)	8.8	9.0	7.1	6.0
Pump max. pressure	(MPa)	35	35		
Pump displacement	(mm^3/rev)	388	386		
Ballscrew gearing	(mm/rev)			19.1	12.0
Gearbox gearing	(-)			47.3	25.5
Actuator characteristics					
Weight	(kg)	9.2	17.2	8.2	13.9
Stall load	(kN)	42.0	42.6	33.6	28.6
Ram speed	(mm/s)	79	78	98	116
Stroke	(mm)	118	117	147	174
Motor characteristics					
Power rating	(kW)	3.7	3.7	3.2	3.2
Corner power	(kW)	12.7	12.7	10.9	11.0
Peak torque	(Nm)	7.9	7.9	7.1	7.1
Continuous torque	(Nm)	3.1	3.1	2.8	2.8

Table 17. Krueger flap actuator

Actuator type →		EHA		EMA	
Configuration →		1EH/SC	2EH/TC	1EM/1BS	2EM/1BS
Design variables					
Kinematic gearing	(rad/m)	6.0	6.0	6.6	6.8
Pump max. pressure	(MPa)	35	35		
Pump displacement	(mm^3/rev)	350	350		
Ballscrew gearing	(mm/rev)			8.2	11.3
Gearbox gearing	(-)			28.9	47.3
Actuator characteristics					
Weight	(kg)	18.2	32.8	11.7	21.0
Stall load	(kN)	42.0	42.0	41.8	47.9
Ram speed	(mm/s)	46	46	42	40
Stroke	(mm)	320	320	293	281
Motor characteristics					
Power rating	(kW)	5.2	5.2	4.7	4.6
Corner power	(kW)	17.8	17.8	16.1	15.8
Peak torque	(Nm)	2.6	2.7	2.8	2.4
Continuous torque	(Nm)	2.6	2.7	2.8	2.4

Table 18. Slat actuator

Actuator type →		EHA		EMA	
Configuration →		1EH/SC	2EH/TC	1EM/1BS	2EM/1BS
Design variables					
Kinematic gearing	(rad/m)	N/A	N/A	N/A	N/A
Pump max. pressure	(MPa)	21	21		
Pump displacement	(mm^3/rev)	300	300		
Ballscrew gearing	(mm/rev)			18.4	22.2
Gearbox gearing	(-)			25.5	30.8
Actuator characteristics					
Weight	(kg)	16.0	27.4	10.6	18.0
Stall load	(kN)	7.8	7.8	7.8	7.8
Ram speed	(mm/s)	60	60	60	60
Stroke	(mm)	663	663	663	663
Motor characteristics					
Power rating	(kW)	1.3	1.3	1.1	1.1
Corner power	(kW)	4.4	4.4	3.8	3.8
Peak torque	(Nm)	3.4	3.4	2.8	2.8
Continuous torque	(Nm)	3.4	3.4	2.8	2.8

Table 19. Trailing edge flap actuator

Actuator type →		EMA		
Load percentage →		50%	70%	100%
Design variables				
Kinematic gearing	(rad/m)	N/A	N/A	N/A
Ballscrew gearing	(mm/rev)	25.1	25.1	25.1
Gearbox gearing	(-)	41.2	41.2	41.2
Actuator characteristics				
Weight	(kg)	33.6	40.6	50.6
Stall load	(kN)	25.5	35.7	51.0
Ram speed	(mm/s)	102	102	102
Motor characteristics				
Power rating	(kW)	6.1	8.5	12.2
Corner power	(kW)	20.7	29.0	41.6
Peak torque	(Nm)	19.8	27.7	39.7
Continuous torque	(Nm)	7.8	10.9	15.6

C. Fault tree analysis for EMA configurations

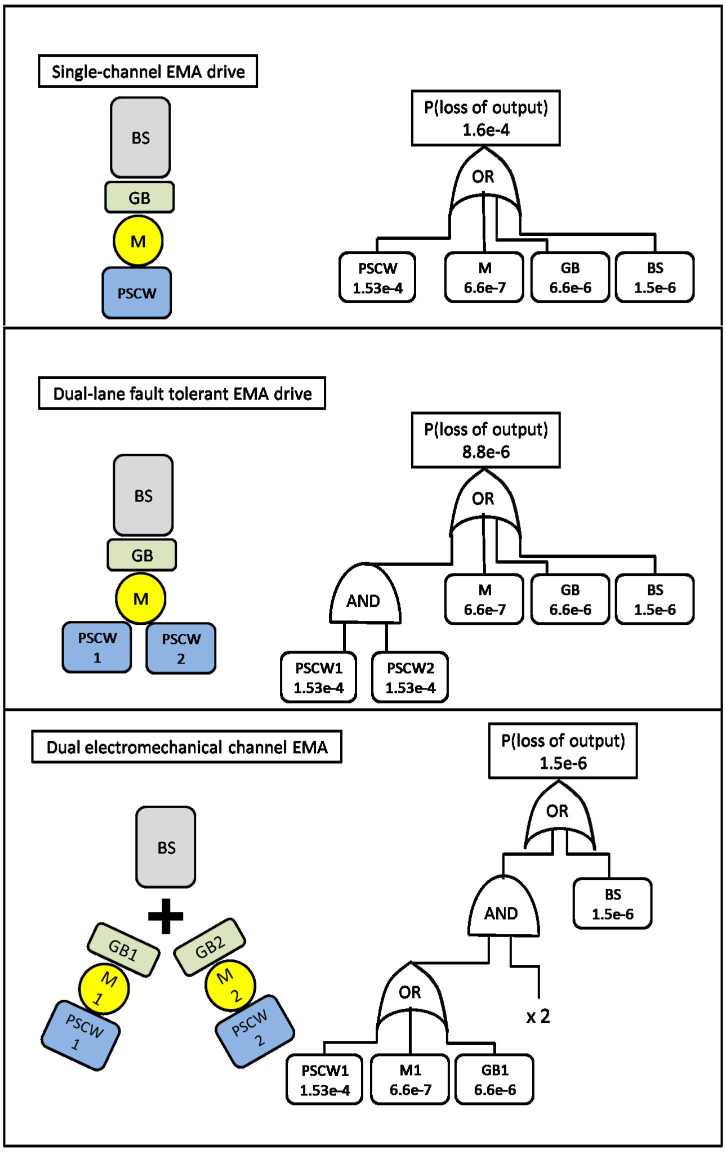


Figure 5. Probability of Loss of Output for EMA configurations (BS - ballscrew, GB - gearbox, M - motor bearings, PSCW - power supply, control, and motor windings)

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